INDOCHAIN GLOBAL MAKMUR

PRODUCT CATALOG 2026



FOUNDRY & CASTING MATERIALS

# FOUNDRY PRODUCTS

Reliable materials for iron, steel and copper-alloy casting.

A SUBSIDIARY OF NCTH

Plus by NCTH NEW CHIEN TE HANG CO.,LTD

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# FOUNDRY PRODUCTS

Reliable foundry materials supporting diverse metal casting processes, from iron and steel to copper alloys and beyond.


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# HIGH STRENGTH PLASTIC

Gray Cast Iron FC,
Ductile Iron FCD,
Stainless Steel, Alloy Steel,
Copper Alloys




We offer a big selection of patching materials for iron casting, steel casting or copper alloy casting.

| Product Name | | Alumina Base | | | | | Silica Base | | |
|---|---|---|---|---|---|---|---|---|---|
| | | R91 | R90 | R71 | R70 | R66S | R81 | R80 | R66 |
| Component | $Al_2O_3$ | >90% | >90% | >70% | >70% | >70% | 5% | 5% | 4% |
| | $SiO_2$ | 5.50% | 5.50% | 9% | 9% | 8% | >90% | >90% | >84% |
| | $Fe_2O_3$ | 1% max | 1% max | 1% max | 1% max | 1% max | 1% max | 1% max | 1% max |
| Density Size | | 2.5t/m³ 0-3mm | 2.5t/m³ 0-6mm | 2t/m³ 0-3mm | 2t/m³ 0-6mm | 2.2t/m³ 0-6mm | 2t/m³ 0-3mm | 2t/m³ 0-6mm | 2t/m³ 0-6mm |
| Max service temperature (°C) | | 1800 | 1800 | 1700 | 1700 | 1600 | 1550 | 1550 | 1550 |
| Application | | Grey Iron / Nodular Iron / Stainless Steel / Alloy Steel / Copper Alloy | | | | Grey Iron / Nodular Iron / Copper Alloy | Grey Iron / Nodular Iron / Copper Alloy | | Grey Iron / Nodular Iron / Copper Alloy |
| | | Ladle / Furnace Lining / Runner / Topping | | | | Ladle / Runner | Ladle / Furnace Lining / Runner / Topping | | Ladle / Runner |

## / DIRECTION /

- Clean and get the construction area wet with water or clay water, and then apply patching materials by hand before smoothing the area with a trowel.
- After construction is done, air dry for 1-3 hours before drying by heat. Then it can be used.

**NOTE** Small cracks are a normal situation, because of water evaporating in the process of air drying and heating. Using finer patching material to repair and drying again will be fine. If the material is too dry to work with, mixing in some water will make it easy to work again.

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# SLAG X

Gray Cast Iron FC, Ductile Iron FCD,
Stainless Steel, Alloy Steel, Copper Alloys



## / DESCRIPTION /

SLAG X is an additive that softens slags and purifies molten metals to help save energy and increase melting efficiency in electric induction furnaces.

### IMMEDIATE BENEFITS

/ Soften the slags
/ Increase the fluidity of molten metals
/ Lessen slags stuck to the furnace lining
/ Keep riser/feeder clean

### SUBSEQUENT BENEFITS

/ Prevent bridging effect
/ Extend furnace lining life by 10% and up
/ Lower pouring temperature
/ Reduce casting defects
/ Enhance casting physical property
/ Improve melting efficiency

## / DIRECTION /

**/ Electric Induction Furnace**
Put SLAG X together with the steel scraps and other materials at the bottom of the furnace.

**/ Ladle**
Put SLAG X at the bottom of the ladle and then pour the molten metals in.

**/ Addition amount**
0.03%-0.08% of molten metals. 0.05% is suggested for initial use. Observe the condition of the furnace lining after each run and then modify the amount accordingly.

**NOTE** Do not apply directly on the surface of the molten metals. Store in a dry atmosphere, otherwise the effectiveness will be affected by moisture. Pay attention to changes of furnace lining as operating conditions vary.

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# EXOTHERMIC ANTI-PIPING COMPOUND

Gray Cast Iron FC, Ductile Iron FCD, Stainless Steel, Alloy Steel



## / DESCRIPTION /

- NCTH Exothermic Anti-piping Compound is used in foundries after pouring, to keep the upper temperature of the molten metals nearly the same as the lower layer, and prevent a piping defect layer from forming.
- Exothermic Anti-Piping Compound keeps the molten metals temperature around the riser/feeder by burning itself to form an insulation layer, ensuring the temperature drops more evenly while the molten metals solidify. The shrinkage shape of the molten metals around the riser/feeder changes from V-shaped to U-shaped.

## / DIRECTION /

- After pouring, apply the compound on the surface of the molten metals through the sprue and riser/feeder head. The compound will react with the molten metals and start burning to supply heat.

## / ADVANTAGES /

- Slow Down Molten Metals Solidifying
- Reduce Heat Loss
- Improved Feeding Capacity
- Decreased Piping Defect
- Few Toxic Smoke or Fumes Emitted

## / SHRINKAGE COMPARISON /



V-shaped U-shaped

**NOTE** Store the compounds in a dry, cool and shady area, away from any heat, flame or anything that sparks.

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# *PLUS+* LOW ASH SEA COAL / COAL DUST

Gray Cast Iron FC,
Ductile Iron FCD



## / FUNCTION /

- **Form Gas Cushion:** To block molten metals from direct contact with the surface of the sand mold and prevent burn-on and burn-in from happening.
- **Offer Reducing Atmosphere:** To reduce the oxidized layer on the molten metal surface and prevent it from being further oxidized, and to prevent burn-on defects caused by reaction between castings and sand on the mold surface.
- **Increase the collapsibility of sand mold:** To make the mold surface smooth and demolding easier, and to reduce burned sand and penetration defects.




▲ Through a controlled and strict manufacturing process, it assures product reliability.

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# SUPER BOND+

Gray Cast Iron FC, Ductile Iron FCD,
Stainless Steel, Alloy Steel, Copper Alloys



## / PRODUCT CHARACTERISTICS /

- Increase green sand strength by more than 20%
- Increase mold surface stability by more than 25% and improve stability of green sand
- Increase the recyclable amount of used sand and decrease the amount of hot sand stuck
- Better filling of detailed parts and corners of the sand mold and lower partial mold failure rate
- Lustrous carbon provides a reducing atmosphere and better air permeability to reduce the castings failure rate and give better precision in size and appearance
- Low water and volatile content; low addition amount
- Better collapsibility to shorten post-processing time







**ADDITION AMOUNT** 0.05-0.08% (Ex: 1000 kg recycled sand, addition amount 0.5-0.8 kg)

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# *PLUS+* BENTOBOND

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## / DESCRIPTION /

Based on artificially sodium-activated bentonite, we have combined a unique non-starch, non-dextrin material carefully formulated for high-temperature conditions. This formula enhances dry compression strength by 7% compared to natural sodium bentonite under severe temperature conditions (above 45°C), while also demonstrating excellent mulling efficiency, allowing you to achieve ideal green strength within effective mixing times.

BentoBond's moderate swelling ensures excellent moldability, effectively reducing mold defects and improving mold quality, with a 7% increase in permeability compared to natural sodium bentonite. With our product, you can achieve higher efficiency and more stable green sand quality.

## / DIRECTION /

Add the amount of existing clay and adjust according to process requirements.

## / ADVANTAGES /

- / Improved clay addition in high sand temperature conditions
- / Excellent moldability and blowing properties
- / Increased mulling efficiency
- / Strengthened dry compression strength
- / Enhanced mold permeability
- / Reduced mold defects

## / EXPERIMENTAL DATA /

| | *PLUS+* BentoBond | Natural Sodium Bentonite |
|---|---|---|
| Moisture (%) | 2.96 | 3.00 |
| Compactability (%) | 39 | 40 |
| Green Compressive Strength (kg/cm²) | 2.09 | 2.08 |
| Permeability | **88** 7% UP | 82 |
| Surface Stability Index (%) | 91.69 | 91.75 |
| Wet Tensile Strength (kg/cm²) | 0.0329 | 0.0351 |
| Dry Compressive Strength (kg/cm²) | **3.26** 7% UP | 3.05 |

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